7.5 The Logistic Model

Name _____ Period ____ Date _____

AP Calculus 7.5 · Student sheet

I can…

- find the carrying capacity from a logistic equation or function.
- find the population when it grows fastest.
- interpret logistic models in context.

Vocabulary · fill in the missing word

Logistic differential equation — $\frac{dP}{dt}$ = _____

Carrying capacity — M, the _____ of P(t) as $t \to \infty$

Logistic function — P(t) = _____

Part 1 · Fish in a pond

A fish population each year.

Year	0	2	4	6	8	10
Fish	50	180	500	820	950	990

Part 2 · Notes and examples (fill in each note, then work the example)

Carrying capacity: M is the _____ of P(t) as $t \to \infty$.

Example 1

$\frac{dP}{dt} = 0.2P(1 - \frac{P}{800})$. Find the carrying capacity.

Fastest growth: the population grows fastest when P = _____.

Example 2

For $\frac{dP}{dt} = 0.2P(1 - \frac{P}{800})$, when is P growing fastest, and how fast?

Logistic function: P(t) = M/(1 + Ae^(−kt)), so P(0) = M/(1 + _____).

Example 3

$P(t) = \frac{600}{1 + 5e^{-0.3t}}$. Find P(0) and the carrying capacity.

Example 4

For $P(t) = \dfrac{600}{1 + 5e^{-0.3t}}$, when is P growing fastest?

Part 3 · Partner practice

Partner 1

$\dfrac{dP}{dt} = 0.0005P(800 - P)$. Find the carrying capacity and P when growth is fastest.

Partner 2

(AP® Practice) A population grows according to $\dfrac{dP}{dt} = 0.01P(3 - \dfrac{P}{2000})$. The carrying capacity is

(A) 2000 (B) 4000 (C) 6000 (D) 10,000